PROBLEM 07

Skater Towed by a Vehicle: Rotating Frame

Original statement · Problems.pdf, p. 8

Rotating Frame + Inertia Forces

A skater B is being towed by a vehicle A on a perfectly smooth horizontal plane, where we define an inertial reference frame Oxy. The vehicle is moving along a circumference of radius a and center at O with constant angular velocity ω. The skater is attached to the vehicle by a bar without mass and constant length $l = a$. Initially, the vehicle is located at $(a, 0)$ and the skater is at rest at $(2a, 0)$. Defining a rotating reference frame Ox_1y_1 with origin at O and axis Ox_1 along OA, and calling ψ the angle between the axis Ox_1 and the bar AB, determine:

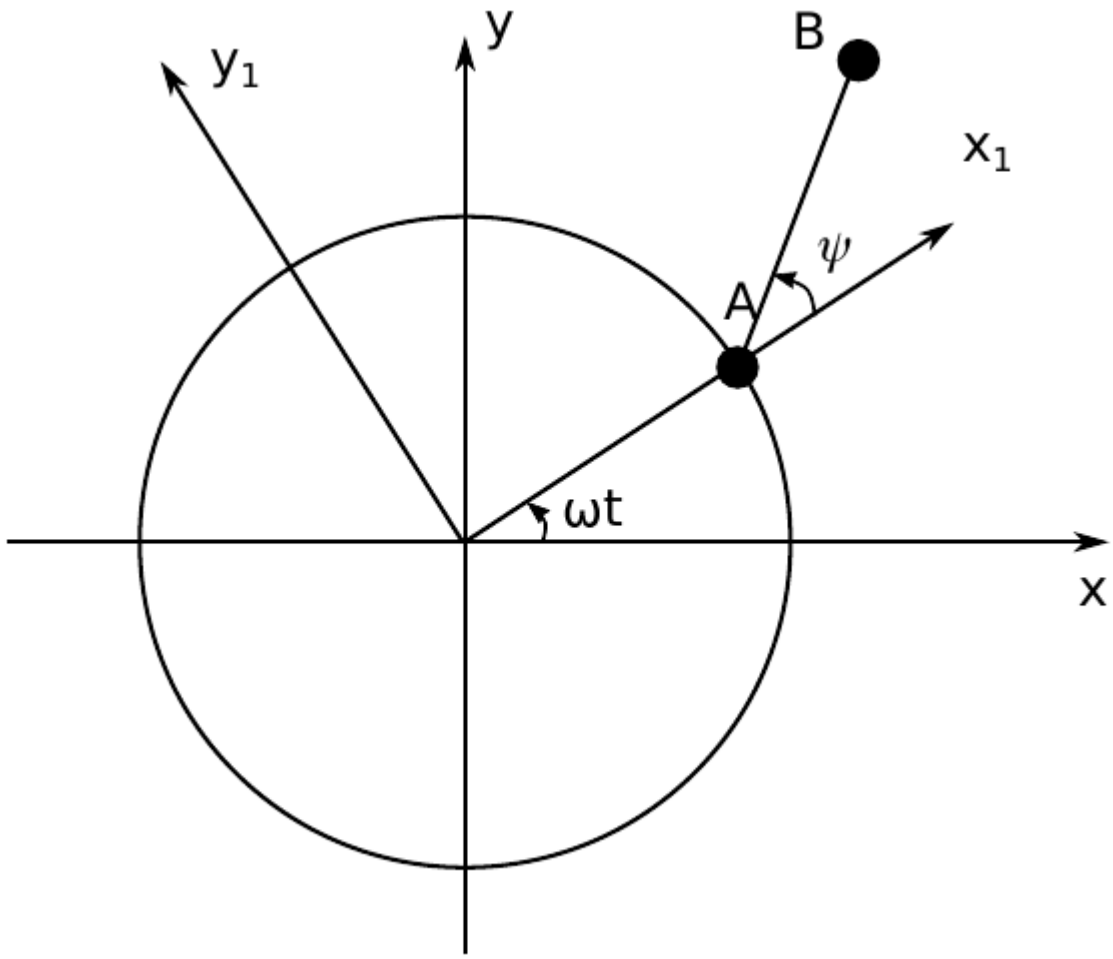


Figure: Inertial axes Oxy, rotating axes Ox1y1 turned by ωt, vehicle A on the circle of radius a, bar AB at angle ψ from Ox1, skater B (Problems.pdf, p. 8).

(a) The relative acceleration of the skater with respect to Ox_1y_1 as a function of ψ and its time derivatives.

(b) The centripetal and Coriolis acceleration terms also as a function of ψ and its time derivatives.

(c) Reduce the evolution of ψ to integrals of the form $\int [\dots]\, d\psi = \int [\dots]\, dt$ (consider that $\dot{\psi}^2 = \int 2\dot{\psi}\ddot{\psi}\, dt$).

(d) Determine the tension on the bar.

(e)

If the rigid bar is substituted by a string which cannot bear compression, determine if/when the string will become loose.

Worked Solution

Reference Frames, Bases and Coordinates

- **Physical system:** the vehicle A moves on the circle of radius a centred at O with constant angular rate ω. The skater B (mass m) is joined to A by a massless rigid bar of length $l = a$ and slides on a smooth horizontal plane.
- **Why a rotating frame.** In the frame Ox_1y_1 that turns with the line OA, the vehicle is at rest at $(a, 0)$, so the skater moves on a fixed circle of radius a around A: one coordinate ψ describes the motion and the constraint is fixed in that frame. The price is that Ox_1y_1 is non-inertial, so we must add the inertia forces of Notes Eq. 4.62 to Newton's law (Notes Eq. 4.63).
- **Hypotheses:**
  1. Oxy is inertial; the plane is smooth (its reaction is vertical and cancels the weight, so neither enters the horizontal dynamics).
  2. The bar is massless, so the force it exerts on B is directed along the bar: $-T\,\mathbf{u}_{AB}$, with $T > 0$ meaning tension.
  3. ω is constant.
- **Fixed (inertial) frame:** $S_0 = Oxy = \{O; \mathbf{i}_0, \mathbf{j}_0, \mathbf{k}_0\}$, basis $\mathcal{B}_0$, with $\mathbf{k}_0$ vertical (normal to the plane of motion).
- **Moving frame:** $S_1 = Ox_1y_1 = \{O; \mathbf{i}_1, \mathbf{j}_1, \mathbf{k}_1\}$, basis $\mathcal{B}_1$, with $\mathbf{i}_1$ along OA and $\mathbf{k}_1 = \mathbf{k}_0$. The origin of S_1 is O (it plays the role of the generic origin "A" of Notes Ch. 3, which here is the fixed point O; the letter A is kept for the vehicle as in the statement) and the particle is $P \equiv B$.
- **Bar basis:** $\{\mathbf{u}_{AB}, \mathbf{u}_\perp, \mathbf{k}_1\}$, the unit vector from A to B and the unit vector normal to the bar in the plane; they are deduced in the next section.
- **Coordinate and degrees of freedom:** B moves in the plane (2 coordinates) with one constraint $|AB| = a$: **1 DOF**, the angle ψ from Ox_1 to the bar AB, measured at A counter-clockwise. The angle ωt of S_1 is prescribed.
- **Initial conditions:** A at $(a, 0)$ and B at rest at $(2a, 0)$, so $\psi(0) = 0$ and $\mathbf{v}_0^B(0) = \mathbf{0}$ (this gives $\dot\psi(0)$ in part (c)).

Symbol	Meaning	SI unit
m	mass of the skater B	kg
a	radius of the vehicle's circle and bar length	m
ω	angular rate of the vehicle (constant)	rad/s
ψ	angle from Ox_1 to the bar AB	rad
T	tension of the bar (positive when pulling B toward A)	N

Change of Basis and Position Vector

The basis $\mathcal{B}_1$ seen from above

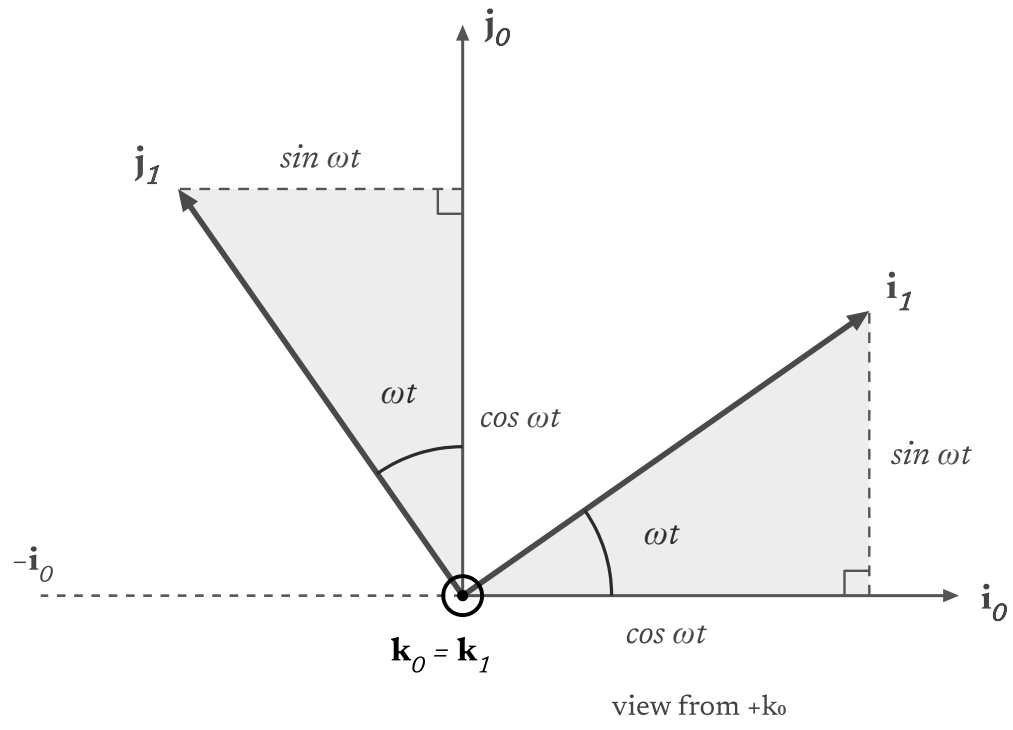


Figure 1. View from $+\mathbf{k}_0$ ($\mathbf{k}_0 = \mathbf{k}_1$ out of the paper, $\odot$). $\mathbf{i}_1$ is $\mathbf{i}_0$ turned counter-clockwise by ωt: legs $\cos\omega t$ along $\mathbf{i}_0$ and $\sin\omega t$ along $\mathbf{j}_0$. $\mathbf{j}_1$ turns the same angle from $\mathbf{j}_0$, so its opposite leg $\sin\omega t$ points along $-\mathbf{i}_0$: hence $\mathbf{j}_1 = -\sin\omega t\,\mathbf{i}_0 + \cos\omega t\,\mathbf{j}_0$.

The rotation is about the vertical $\mathbf{k}_0$, so it is drawn looking down from $+\mathbf{k}_0$: $\mathbf{k}_0 = \mathbf{k}_1$ points out of the paper, towards the reader ($\odot$). With $\mathbf{i}_0$ to the right, the right-hand rule $\mathbf{k}_0 = \mathbf{i}_0 \times \mathbf{j}_0$ puts $\mathbf{j}_0$ up the page (fingers from $\mathbf{i}_0$ to $\mathbf{j}_0$, counter-clockwise, thumb towards the reader). The

vehicle starts on Ox and turns counter-clockwise, so after a time t the angle from $\mathbf{i}_0$ to $\mathbf{i}_1$ is ωt :

- $\mathbf{i}_1$ is the hypotenuse (length 1) of a right triangle whose leg adjacent to ωt lies along $\mathbf{i}_0$ and measures $\cos\omega t$, and whose opposite leg lies along $+\mathbf{j}_0$ and measures $\sin\omega t$: $\mathbf{i}_1 = \cos\omega t\,\mathbf{i}_0 + \sin\omega t\,\mathbf{j}_0$.
- $\mathbf{j}_1$ is $\mathbf{i}_1$ turned a further 90° counter-clockwise (so that $\mathbf{k}_1 = \mathbf{i}_1 \times \mathbf{j}_1$ still points towards the reader). It has turned the same angle ωt from $\mathbf{j}_0$, and turning counter-clockwise from "up" moves its tip to the left: its leg along $\mathbf{j}_0$ (adjacent) is $\cos\omega t$ and its leg along $-\mathbf{i}_0$ (opposite) is $\sin\omega t$. **That is the origin of the minus sign:** $\mathbf{j}_1 = -\sin\omega t\,\mathbf{i}_0 + \cos\omega t\,\mathbf{j}_0$.
- $\mathbf{k}_1 = \mathbf{k}_0$, the axis of the rotation.

The columns of $[{}_0R_1]$ are the components of $\mathbf{i}_1, \mathbf{j}_1, \mathbf{k}_1$ in $\mathcal{B}_0$:

$$[{}_0R_1] = \begin{pmatrix} \cos\omega t & -\sin\omega t & 0 \\ \sin\omega t & \cos\omega t & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

Checks: $\det[{}_0R_1] = \cos\omega t\cos\omega t - (-\sin\omega t)\sin\omega t = \cos^2\omega t + \sin^2\omega t = 1$; the columns are unit vectors ($\cos^2\omega t + \sin^2\omega t = 1$) and mutually orthogonal ($-\cos\omega t\sin\omega t + \sin\omega t\cos\omega t = 0$), so $[{}_0R_1][{}_0R_1]^T = I$.

Motion of S_1. Simple rotation about $\mathbf{k}_0 = \mathbf{k}_1$ (Slide 16):

$$\boldsymbol{\omega}_{10} = \omega\,\mathbf{k}_1, \qquad \boldsymbol{\alpha}_{10} = \left.\frac{d\boldsymbol{\omega}_{10}}{dt}\right|_0 = \dot\omega\,\mathbf{k}_1 = \mathbf{0}, \qquad \mathbf{r}_0^O = \mathbf{v}_0^O = \mathbf{a}_0^O = \mathbf{0}$$

Check with Poisson's formula: by the chain rule $\frac{d\mathbf{i}_1}{dt}\big|_0 = -\omega \sin\omega t\, \mathbf{i}_0 + \omega\cos\omega t\, \mathbf{j}_0 = \omega\, \mathbf{j}_1$, and $\boldsymbol{\omega}_{10} \times \mathbf{i}_1 = \omega\, \mathbf{k}_1 \times \mathbf{i}_1 = \omega\, \mathbf{j}_1$.

The position vector in the rotating plane

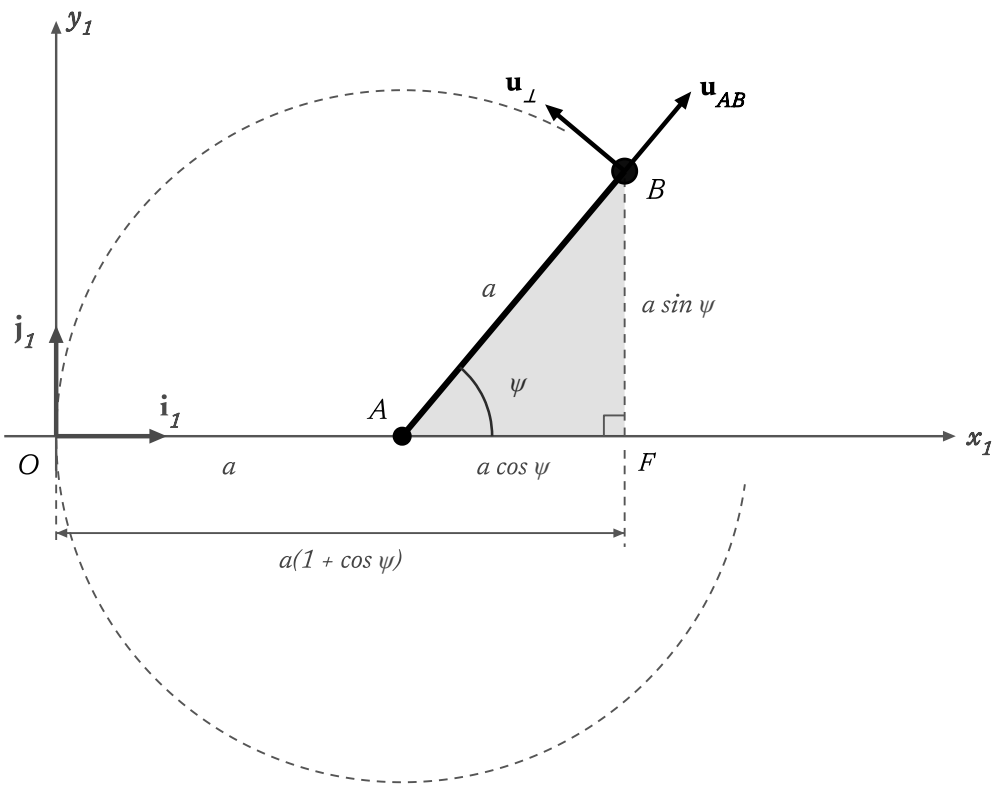


Figure 2. Plane of S_1 ($\mathbf{i}_1$ along OA, $\mathbf{k}_1$ out of the paper). $\mathbf{r}_1^B = OA + AB$: in the shaded right triangle AFB the hypotenuse is $|AB| = a$, the leg adjacent to ψ is $a\cos\psi$ along $\mathbf{i}_1$ and the opposite leg is $a\sin\psi$ along $\mathbf{j}_1$. The dashed circle is the path of B relative to S_1; $\mathbf{u}_{AB}, \mathbf{u}_\perp$ in orange.

In S_1 the vehicle is at rest at $A = (a, 0)$, so the plane is drawn with $\mathbf{i}_1$ to the right and $\mathbf{j}_1$ up. To go from O to B we pass through A, because the distance from A is the one fixed by the bar (Notes Eq. 3.16 with $\mathbf{r}_0^O = \mathbf{0}$):

$$\mathbf{r}_1^B = \overrightarrow{OA} + \overrightarrow{AB}$$

- $\overrightarrow{OA} = a\,\mathbf{i}_1$: A lies on Ox_1 at distance a from O.
- $\overrightarrow{AB}$ has length a and makes the angle ψ with $\mathbf{i}_1$, measured counter-clockwise at A. Let F be the foot of the perpendicular from B to the line Ox_1. In the shaded right triangle AFB (right angle at F) the hypotenuse is $|AB| = a$, the leg adjacent to ψ is $AF = a\cos\psi$

along $\mathbf{i}_1$ and the leg opposite to ψ is $FB = a\sin\psi$ along $\mathbf{j}_1$. So $\overrightarrow{AB} = a\cos\psi\,\mathbf{i}_1 + a\sin\psi\,\mathbf{j}_1$.

Adding the two vectors:

$$\mathbf{r}_1^B = a\,\mathbf{i}_1 + a\left(\cos\psi\,\mathbf{i}_1 + \sin\psi\,\mathbf{j}_1\right) = a(1+\cos\psi)\,\mathbf{i}_1 + a\sin\psi\,\mathbf{j}_1$$

These are the dashed components of the figure: $|OF| = a + a\cos\psi$ along $\mathbf{i}_1$ and $|FB| = a\sin\psi$ along $\mathbf{j}_1$. For $\psi < 0$ (the case of the motion, as part (c) shows) $\sin\psi < 0$ and B is below Ox_1; the same formula holds. Here ψ is the standard polar angle of B about A (measured from $\mathbf{i}_1$, counter-clockwise), so no shift of the angle is needed. Checks: $\psi = 0$ gives $2a\,\mathbf{i}_1$, the initial position; $\psi = -\pi$ gives $\mathbf{0}$, the point O.

Bar basis. The unit vector along the bar is $\mathbf{u}_{AB} = \overrightarrow{AB}/a = \cos\psi\,\mathbf{i}_1 + \sin\psi\,\mathbf{j}_1$ (from A to B). Differentiating with respect to ψ ($\frac{d}{d\psi}\cos\psi = -\sin\psi$, $\frac{d}{d\psi}\sin\psi = \cos\psi$) gives the unit vector normal to the bar, in the sense of increasing ψ:

$$\mathbf{u}_\perp = \frac{d\mathbf{u}_{AB}}{d\psi} = -\sin\psi\,\mathbf{i}_1 + \cos\psi\,\mathbf{j}_1 = \mathbf{k}_1 \times \mathbf{u}_{AB}$$

(the last form with $\mathbf{k}_1 \times \mathbf{i}_1 = \mathbf{j}_1$ and $\mathbf{k}_1 \times \mathbf{j}_1 = -\mathbf{i}_1$). Their matrix $\begin{pmatrix} \cos\psi & -\sin\psi \\ \sin\psi & \cos\psi \end{pmatrix}$ (completed with $\mathbf{k}_1$) is again a simple rotation: $\det = \cos^2\psi + \sin^2\psi = 1$, and RR^T has diagonal entries $\cos^2\psi + \sin^2\psi = 1$ and off-diagonal entries $\cos\psi\sin\psi - \sin\psi\cos\psi = 0$, so $RR^T = I$. Also $\frac{d\mathbf{u}_\perp}{d\psi} = -\cos\psi\,\mathbf{i}_1 - \sin\psi\,\mathbf{j}_1 = -\mathbf{u}_{AB}$.

Kinematics

(a) Relative acceleration $\mathbf{a}_1^B$

> **Pedagogical Justification:** the relative quantities are derivatives in S_1, where $\mathbf{i}_1, \mathbf{j}_1$ are constant, so only the components are differentiated.

Relative velocity, with the chain rule $\frac{d}{dt}\cos\psi = -\sin\psi\,\dot\psi$ and $\frac{d}{dt}\sin\psi = \cos\psi\,\dot\psi$:

$$\mathbf{v}_1^B = \left.\frac{d\mathbf{r}_1^B}{dt}\right|_1 = -a\dot{\psi}\sin\psi\,\mathbf{i}_1 + a\dot{\psi}\cos\psi\,\mathbf{j}_1 = a\dot{\psi}\,\mathbf{u}_\perp$$

Relative acceleration, with the product rule $\frac{d}{dt}(\dot{\psi}\sin\psi) = \ddot{\psi}\sin\psi + \dot{\psi}^2\cos\psi$ and $\frac{d}{dt}(\dot{\psi}\cos\psi) = \ddot{\psi}\cos\psi - \dot{\psi}^2\sin\psi$:

$$\boxed{\mathbf{a}_1^B = -a\left(\ddot{\psi}\sin\psi + \dot{\psi}^2\cos\psi\right)\mathbf{i}_1 + a\left(\ddot{\psi}\cos\psi - \dot{\psi}^2\sin\psi\right)\mathbf{j}_1 = a\ddot{\psi}\,\mathbf{u}_\perp - a\dot{\psi}^2\,\mathbf{u}_{AB}}$$

(the last form follows by grouping the $\ddot{\psi}$ and $\dot{\psi}^2$ terms: circular motion of radius a about A, tangential plus normal part).

(b) Centripetal and Coriolis terms

> **Pedagogical Justification:** they are the two non-zero rotation terms of Notes Eq. 3.27 (Slide 25); the Euler term vanishes because $\boldsymbol{\alpha}_{10} = \mathbf{0}$ and the origin term because $\mathbf{a}_0^O = \mathbf{0}$.

Cross products used: $\mathbf{k}_1 \times \mathbf{i}_1 = \mathbf{j}_1$, $\mathbf{k}_1 \times \mathbf{j}_1 = -\mathbf{i}_1$.

Centripetal term.

$$\boldsymbol{\omega}_{10} \times \mathbf{r}_1^B = \omega\mathbf{k}_1 \times [a(1+\cos\psi)\mathbf{i}_1 + a\sin\psi\,\mathbf{j}_1] = a\omega(1+\cos\psi)\,\mathbf{j}_1 - a\omega\sin\psi\,\mathbf{i}_1$$

$$\boldsymbol{\omega}_{10} \times \left(\boldsymbol{\omega}_{10} \times \mathbf{r}_1^B\right) = \omega\mathbf{k}_1 \times [-a\omega\sin\psi\,\mathbf{i}_1 + a\omega(1+\cos\psi)\,\mathbf{j}_1] = -a\omega^2\sin\psi\,\mathbf{j}_1 - a\omega^2(1+\cos\psi)\,\mathbf{i}_1$$

$$\boxed{\boldsymbol{\omega}_{10} \times \left(\boldsymbol{\omega}_{10} \times \mathbf{r}_1^B\right) = -a\omega^2\left[(1+\cos\psi)\,\mathbf{i}_1 + \sin\psi\,\mathbf{j}_1\right] = -\omega^2\,\mathbf{r}_1^B}$$

It equals $-\omega^2\mathbf{r}_\perp$ with $\mathbf{r}_\perp = \mathbf{r}_1^B$ because $\mathbf{r}_1^B$ lies in the plane normal to $\boldsymbol{\omega}_{10}$ (Slide 30).

Coriolis term.

$$2\boldsymbol{\omega}_{10}\times\mathbf{v}_1^B = 2\omega\mathbf{k}_1\times\left(-a\dot{\psi}\sin\psi\,\mathbf{i}_1 + a\dot{\psi}\cos\psi\,\mathbf{j}_1\right) = -2a\omega\dot{\psi}\sin\psi\,\mathbf{j}_1 - 2a\omega\dot{\psi}\cos\psi\,\mathbf{i}_1$$

$$\boxed{2\boldsymbol{\omega}_{10}\times\mathbf{v}_1^B = -2a\omega\dot{\psi}\left(\cos\psi\,\mathbf{i}_1 + \sin\psi\,\mathbf{j}_1\right) = -2a\omega\dot{\psi}\,\mathbf{u}_{AB}}$$

Absolute acceleration (Notes Eq. 3.27):

$$\mathbf{a}_0^B = \underbrace{\mathbf{a}_0^O}_{\mathbf{0}} + \mathbf{a}_1^B + \underbrace{\boldsymbol{\alpha}_{10}\times\mathbf{r}_1^B}_{\mathbf{0}} - \omega^2\mathbf{r}_1^B - 2a\omega\dot{\psi}\,\mathbf{u}_{AB}$$

Forces and Equations of Motion

(c) Forces on the skater in S_1

Pedagogical Justification: in S_1 Newton's law holds with the inertia forces (Notes Eqs. 4.62–4.63): $m\mathbf{a}_1^B = \mathbf{F} + \mathbf{F}_i$. Here $\mathbf{F} = -T\,\mathbf{u}_{AB}$ (horizontal part), $\mathbf{F}_A = \mathbf{0}$, $\mathbf{F}_E = \mathbf{0}$, and

$$\mathbf{F}_C = -m\boldsymbol{\omega}_{10}\times\left(\boldsymbol{\omega}_{10}\times\mathbf{r}_1^B\right) = ma\omega^2\left[(1+\cos\psi)\,\mathbf{i}_1 + \sin\psi\,\mathbf{j}_1\right], \qquad \mathbf{F}_{COR} = -2m\boldsymbol{\omega}_{10}\times\mathbf{v}_1^B = 2ma\omega\dot{\psi}\,\mathbf{u}_{AB}$$

(c, continued) Equation of motion

The unknown T disappears if we project on $\mathbf{u}_\perp$, the only direction along which the bar exerts no force.

Projections on $\mathbf{u}_\perp = -\sin\psi\,\mathbf{i}_1 + \cos\psi\,\mathbf{j}_1$:

- $m\mathbf{a}_1^B\cdot\mathbf{u}_\perp = ma\ddot{\psi}$ (from the grouped form of part (a), since $\mathbf{u}_{AB}\cdot\mathbf{u}_\perp = 0$);
- $-T\,\mathbf{u}_{AB}\cdot\mathbf{u}_\perp = 0$ and $\mathbf{F}_{COR}\cdot\mathbf{u}_\perp = 0$;
- $\mathbf{F}_C\cdot\mathbf{u}_\perp = ma\omega^2\left[-(1+\cos\psi)\sin\psi + \sin\psi\cos\psi\right] = -ma\omega^2\sin\psi$.

$$ma\ddot{\psi} = -ma\omega^2 \sin\psi \quad \Longrightarrow \quad \boxed{\ddot{\psi} + \omega^2 \sin\psi = 0}$$

In S_1 the skater behaves as a pendulum of length a hanging from A: about A the centrifugal force $m\omega^2(\mathbf{OA} + \mathbf{AB})$ has the same moment as the uniform field $ma\omega^2\,\mathbf{i}_1$ (because $\mathbf{AB} \times \mathbf{AB} = \mathbf{0}$), so the "effective gravity" has strength $a\omega^2$ and points along $+\mathbf{i}_1$.

(d) Equation for the tension

> **Pedagogical Justification:** project $m\mathbf{a}_1^B = -T\mathbf{u}_{AB} + \mathbf{F}_C + \mathbf{F}_{COR}$ on $\mathbf{u}_{AB}$, the direction in which the bar force acts.

Projections on $\mathbf{u}_{AB} = \cos\psi\,\mathbf{i}_1 + \sin\psi\,\mathbf{j}_1$:

- $m\mathbf{a}_1^B \cdot \mathbf{u}_{AB} = -ma\dot{\psi}^2$;
- $\mathbf{F}_C \cdot \mathbf{u}_{AB} = ma\omega^2\left[(1+\cos\psi)\cos\psi + \sin^2\psi\right] = ma\omega^2(1+\cos\psi)$;
- $\mathbf{F}_{COR} \cdot \mathbf{u}_{AB} = 2ma\omega\dot{\psi}$.

$$-ma\dot{\psi}^2 = -T + ma\omega^2(1+\cos\psi) + 2ma\omega\dot{\psi} \quad \Longrightarrow \quad T = ma\left[\dot{\psi}^2 + 2\omega\dot{\psi} + \omega^2(1+\cos\psi)\right]$$

Answers and Checks

(a) Relative acceleration

$\mathbf{a}_1^B = -a\left(\ddot{\psi}\sin\psi + \dot{\psi}^2\cos\psi\right)\mathbf{i}_1 + a\left(\ddot{\psi}\cos\psi - \dot{\psi}^2\sin\psi\right)\mathbf{j}_1 = a\ddot{\psi}\,\mathbf{u}_\perp - a\dot{\psi}^2\,\mathbf{u}_{AB}$.

(b) Centripetal and Coriolis terms

$\boldsymbol{\omega}_{10} \times \left(\boldsymbol{\omega}_{10} \times \mathbf{r}_1^B\right) = -a\omega^2\left[(1+\cos\psi)\,\mathbf{i}_1 + \sin\psi\,\mathbf{j}_1\right]$ and $2\boldsymbol{\omega}_{10} \times \mathbf{v}_1^B = -2a\omega\dot{\psi}\,(\cos\psi\,\mathbf{i}_1 + \sin\psi\,\mathbf{j}_1)$. Both agree with the official key (Problems.pdf, p. 72), where the centripetal term is called "centrifugal acceleration".

(c) First integral, initial rate and quadrature

First integral (hint of the statement). Multiply $\ddot{\psi} + \omega^2 \sin\psi = 0$ by $2\dot{\psi}$. By the chain rule $\frac{d}{dt}\left(\dot{\psi}^2\right) = 2\dot{\psi}\ddot{\psi}$ and $\frac{d}{dt}\cos\psi = -\sin\psi\,\dot{\psi}$, so

$$2\dot{\psi}\ddot{\psi} = -2\omega^2 \sin\psi\,\dot{\psi} \quad\Longleftrightarrow\quad \frac{d}{dt}\left(\dot{\psi}^2\right) = 2\omega^2 \frac{d}{dt}\left(\cos\psi\right)$$

Integrate from 0 to t and apply Barrow's rule $[F]_0^t = F(t) - F(0)$:

On the left, $\int_0^t 2\dot{\psi}\ddot{\psi}\,dt' = \left[\dot{\psi}^2\right]_0^t$. On the right, the change of variable $\psi = \psi(t')$, $d\psi = \dot{\psi}\,dt'$, turns $\int_0^t \left(-2\omega^2\sin\psi\right)\dot{\psi}\,dt'$ into $\int_{\psi(0)}^{\psi(t)} \left(-2\omega^2\sin\psi\right)d\psi$, whose antiderivative is $2\omega^2\cos\psi$:

$$\dot{\psi}^2(t) - \dot{\psi}^2(0) = \int_{\psi(0)}^{\psi(t)} \left(-2\omega^2\sin\psi\right)d\psi = 2\omega^2\left[\cos\psi\right]_{\psi(0)}^{\psi(t)} = 2\omega^2\left(\cos\psi - 1\right)$$

Initial rate. By Notes Eq. 3.26, $\mathbf{v}_0^B = \mathbf{v}_0^O + \mathbf{v}_1^B + \boldsymbol{\omega}_{10}\times\mathbf{r}_1^B$. At $t=0$: $\psi = 0$, $\mathbf{r}_1^B = 2a\,\mathbf{i}_1$, $\mathbf{v}_1^B = a\dot{\psi}(0)\,\mathbf{j}_1$, $\boldsymbol{\omega}_{10}\times\mathbf{r}_1^B = 2a\omega\,\mathbf{j}_1$, and $\mathbf{v}_0^B = \mathbf{0}$:

$$\mathbf{0} = \Big[a\dot{\psi}(0) + 2a\omega\Big]\,\mathbf{j}_1 \quad\Longrightarrow\quad \dot{\psi}(0) = -2\omega$$

Hence, with $1 + \cos\psi = 2\cos^2(\psi/2)$,

$$\dot{\psi}^2 = 4\omega^2 + 2\omega^2(\cos\psi - 1) = 2\omega^2(1+\cos\psi) = 4\omega^2\cos^2\frac{\psi}{2}$$

For $-\pi < \psi < \pi$ the factor $\cos(\psi/2) > 0$, so $\dot{\psi}$ cannot vanish nor change sign there; $\dot{\psi}(0) = -2\omega < 0$ selects

$$\boxed{\dot{\psi} = -2\omega\cos\frac{\psi}{2}, \qquad -\pi < \psi \le 0}$$

which agrees with the official key (Problems.pdf, p. 72).

Energy check in S_1 (Slides 29–30). The origin is fixed and $\boldsymbol{\omega}_{10}$ is constant, so $\mathbf{F}_C$ has the potential $V_1 = -\tfrac{1}{2}m\omega^2 r_\perp^2$, $\mathbf{F}_{COR}$ does no work, and the bar force does no work in S_1 ($-T\mathbf{u}_{AB}\cdot a\dot\psi\mathbf{u}_\perp = 0$). With $r_\perp^2 = a^2\left[(1+\cos\psi)^2+\sin^2\psi\right] = 2a^2(1+\cos\psi)$:

$$E_1^B = \frac{1}{2}ma^2\dot\psi^2 - ma^2\omega^2(1+\cos\psi) = \text{const} = \frac{1}{2}ma^2(4\omega^2) - 2ma^2\omega^2 = 0$$

which gives again $\dot\psi^2 = 2\omega^2(1+\cos\psi)$.

Quadrature. Separate variables in $\frac{d\psi}{dt} = -2\omega\cos\frac{\psi}{2}$:

$$\boxed{\int_0^{\psi}\frac{d\psi'}{2\cos(\psi'/2)} = -\int_0^t \omega\,dt'}$$

The left integral is elementary. Substitute $w = \psi'/2$, $d\psi' = 2\,dw$, limits $0\to\psi/2$; the antiderivative of $\sec w$ is $\ln\left|\sec w+\tan w\right|$ (its derivative is $\frac{\sec w\tan w+\sec^2 w}{\sec w+\tan w} = \sec w$). By Barrow's rule:

$$\int_0^{\psi/2}\frac{dw}{\cos w} = \left[\ln\left|\sec w+\tan w\right|\right]_0^{\psi/2} = \ln\left|\sec\frac{\psi}{2}+\tan\frac{\psi}{2}\right| - \ln 1 = -\omega t$$

so $\psi(t)$ is given implicitly by $\sec\frac{\psi}{2}+\tan\frac{\psi}{2} = e^{-\omega t}$; as $t\to\infty$, $\psi\to-\pi$, i.e. B tends to the centre O ($\mathbf{r}_1^B\to\mathbf{0}$) without ever reaching it.

(d) Tension on the bar

Substitute the first integral in $T = ma\left[\dot\psi^2 + 2\omega\dot\psi + \omega^2(1+\cos\psi)\right]$, with $c\equiv\cos(\psi/2)$: $\dot\psi^2 = 4\omega^2c^2$, $\dot\psi = -2\omega c$, $1+\cos\psi = 2c^2$:

$$T = ma\omega^2\left(4c^2 - 4c + 2c^2\right) = ma\omega^2\left(6c^2-4c\right)$$

$$\boxed{T = 2ma\omega^2 \cos\frac{\psi}{2}\left(3\cos\frac{\psi}{2} - 2\right)}$$

which agrees with the official key (Problems.pdf, p. 72). At $t = 0$: $T(0) = 2ma\omega^2 > 0$.

(e) Does a string become loose?

> **Pedagogical Justification:** a string can only pull, so it stays taut while $T \geq 0$; it goes slack at the first instant where T would become negative.

As ψ decreases from 0 toward $-\pi$, $c = \cos(\psi/2)$ decreases monotonically from 1 toward 0. The factor c stays positive, and $3c - 2$ changes sign at $c = 2/3$:

$$\cos\frac{\psi}{2} = \frac{2}{3} \quad\Longrightarrow\quad \boxed{\psi_{\text{loose}} = -2\arccos\frac{2}{3} \approx -1.682\ \text{rad} \approx -96.38^\circ}$$

(official key: $\psi = -2\arccos(2/3)$). **When:** at $\psi/2 = -\arccos(2/3)$, $\sec(\psi/2) = 3/2$ and $\tan(\psi/2) = -\sqrt{1 - 4/9}/(2/3) = -\sqrt{5}/2$, so from part (c)

$$\omega t_{\text{loose}} = -\ln\left(\frac{3}{2} - \frac{\sqrt{5}}{2}\right) = \ln\frac{2}{3 - \sqrt{5}} = \ln\frac{3 + \sqrt{5}}{2} \approx 0.9624, \qquad t_{\text{loose}} \approx \frac{0.962}{\omega}$$

So yes: the string goes slack after about $0.96/\omega$ seconds, when the bar has turned 96.4° clockwise from Ox_1.

Checks

- **Reading in S_1.** The rotating observer sees B as a pendulum of length a hung from A in an effective field of strength $a\omega^2$ pointing away from O (the centrifugal force), whose "lowest" point is $\psi = 0$. The skater starts there with relative speed $2a\omega$, which is exactly the separatrix energy ($E_1^B = 0$): it creeps toward the "inverted" position $\psi = -\pi$ (where B would sit at O) in infinite time.
- **Why T drops.** The Coriolis force $2ma\omega\dot{\psi}\,\mathbf{u}_{AB}$ points toward A when $\dot{\psi} < 0$, and the centrifugal pull along the bar, $ma\omega^2(1 + \cos\psi)$, shrinks as B swings in; when the required centripetal force $ma\dot{\psi}^2$ is supplied by these inertia forces alone the bar is unloaded ($T = 0$).

- **Limit $\omega \to 0$:** $\dot\psi \to 0, T \to 0$ (nothing moves).
- **Orthogonality:** $\mathbf{u}_{AB} \cdot \mathbf{u}_\perp = -\cos\psi\sin\psi + \sin\psi\cos\psi = 0$, $|\mathbf{u}_{AB}| = |\mathbf{u}_\perp| = 1$.
- **Dimensions:** $[\mathbf{a}_1^B] = [a\ddot\psi] = [a\dot\psi^2] = \mathrm{m\,s^{-2}}$; $[\omega^2 a] = [\omega\dot\psi a] = \mathrm{m\,s^{-2}}$; $[T] = [ma\omega^2] = \mathrm{kg\,m\,s^{-2}} = \mathrm{N}$; $[\omega t] = 1$.
- **Verification (sympy):** the inertial double derivative of $[{}_0R_1]\{\mathbf{r}_1^B\}$ equals $\mathbf{a}_1^B - \omega^2\mathbf{r}_1^B - 2a\omega\dot\psi\mathbf{u}_{AB}$; the EOM, $\dot\psi(0) = -2\omega$, $dE_1^B/dt = 0$ along the EOM, the tension and ψ_{loose} are reproduced; an RK4 integration with $\omega = 1$ reaches $\cos(\psi/2) = 2/3$ at $t = 0.9625$ s, matching $\ln\frac{3+\sqrt{5}}{2} = 0.9624$.
- **Check against the official key:** the key of `Problems.pdf` (p. 72) gives the centripetal and Coriolis terms, $\dot\psi = -2\omega\cos(\psi/2)$, $T = 2ma\omega^2\cos(\psi/2)\,(3\cos(\psi/2) - 2)$ and $\psi = -2\arccos(2/3)$ for the string going slack. All of them coincide with the results above (no discrepancy).

Source: Problems.pdf (PDF page 8); key on PDF page 72